Grade: Check Plus

Statistics and Data Analysis: Expected Values and Options

Based on the provided probability distribution for the stock price (x) one year from now:

Price (x)	**Probability f(x)**
20	.05
21	.06
22	.07
23	.08
24	.09
25	.12
26	.15
27	.13
28	.10

29	.10
30	.05

1. Mean and Standard Deviation of the Stock Price

To calculate the mean (μ) and standard deviation (σ) of the stock price:

Expected Value (Mean) E(X)

The mean is calculated as $\sum x \cdot f(x)$:

- 20(.05)+21(.06)+22(.07)+23(.08)+24(.09)+25(.12)+26(.15)+27(.13)+28(.10)+29(.10)+30(.05)
- 1.00+1.26+1.54+1.84+2.16+3.00+3.90+3.51+2.80+2.90+1.50=25.41

Standard Deviation σ

First, calculate $E(X_2)=\sum x_2 \cdot f(x)$:

- 400(.05)+441(.06)+484(.07)+529(.08)+576(.09)+625(.12)+676(.15)+729(.13)+784(.10)+841(.10)+900(.05)
- 20+26.46+33.88+42.32+51.84+75+101.4+94.77+78.4+84.1+45=653.17

$Var(X)=E(X_2)-[E(X)]_2=653.17-(25.41)_2=653.17-645.6681=7.5019$
$\sigma \approx 2.74$

✓ Q1

2. Probabilities within Standard Deviations

a. Within 1 Standard Deviation

The range is [22.67, 28.15]. The prices falling in this range are 23, 24, 25, 26, 27, and 28.

✓ Q2a

$P(22.67 \leq X \leq 28.15)=.08+.09+.12+.15+.13+.10=0.67$

b. Within 2 Standard Deviations

The range is [19.93, 30.89]. All prices in the table (20 through 30) fall within this range.

$P(19.93 \leq X \leq 30.89)=1.00$ ✓ Q2b

3. Distribution of the Option Value

The option allows the purchase of stock at $25. The value V is max(0,x−25):

Stock Price (x)	Option Value (v)	Probability f(v)
20, 21, 22, 23, 24, 25	0	.05+.06+.07+.08+.09+.12 = **.47**
26	1	**.15**
27	2	**.13**
28	3	**.10**
29	4	**.10**
30	5	**.05**

✓ Q3

4. Probability the Option has Positive Value

The option has positive value when the stock price is greater than $25 (i.e., x=26,27,28,29,30).

P(V>0)=.15+.13+.10+.10+.05=0.53 ✓ Q4

5. Mean and Standard Deviation of the Option Value

Expected Value E(V)

E(V)=0(.47)+1(.15)+2(.13)+3(.10)+4(.10)+5(.05)
E(V)=0+.15+.26+.30+.40+.25=1.36 ✓ Q5

Standard Deviation of Option Value

First, $E(V2)=0^2(.47)+1^2(.15)+2^2(.13)+3^2(.10)+4^2(.10)+5^2(.05)$:

- 0+.15+.52+.90+1.60+1.25=4.42

$Var(V)=4.42-(1.36)^2=4.42-1.8496=2.5704$
$\sigma V\approx 1.60$